

Carole R. Hall

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EDUCATION:

- **Stony Brook University**, Stony Brook, NY

Applied Mathematics and Statistics:

August 2020 - May 2026

Computational Biology, PhD.

Dissertation Title: "Geometric Methods for the Analysis of Remote Sensing Data"

Advisor: Joseph Mitchell, Distinguished Professor in Applied Mathematics and Statistics

- **University of Minnesota, Twin Cities**, Minneapolis, MN

Mathematical Biology: Genomics, B.S.

September 2016 - May 2020

SKILLS:

- Python (Scikit-Learn, Scipy, PyTorch, Keras, Tensorflow, Geopandas), R, MATLAB, C++, SQL, XML.
- Machine learning, deep learning, spatial statistics, time-series modeling, anomaly detection, GIS.
- Git, high-performance computing (HPC).
- Image processing, computer vision, object detection, functional & shape data analysis.
- Excellent written & oral communication skills, LaTeX document preparation.

WORK & RESEARCH EXPERIENCE:

- **The Colorado School of Mines**, Golden, CO

NIST PREP Post-Doctoral Fellow

June 2026 - Present

- Work in a joint program between NIST and Mines under Prof. Greg Fasshauer funded by CHIPS for America.
- Develop shape analysis methods to improve the manufacturing of silicon computer chips.

- **Sandia National Laboratories**, Albuquerque, NM

Statistical Sciences PhD Technical Intern

June 2023 - Present

- Researched analytical methods in statistical shape analysis & computer vision.
- Created forecasting algorithms for large sets of spatially-dependent climate data.
- Implemented machine learning & deep learning techniques on image data.
- Processed dynamic aerial imagery & perform subspace tracking & object detection.
- Performed shape analysis on spectral signature data & SEM imagery of material particles.
- Collaborated with researchers from other national labs on security & scientific problems.
- Developed software in R, Python, MATLAB; configure XML-dependent pipelines.

- **Stony Brook University**, Stony Brook, NY

Research Assistant to Professor Joseph Mitchell and Heather Lynch

August 2020 - May 2026

- Performed NASA-funded research spanning across multiple disciplines.
- Processed satellite imagery & drone imagery over high & low resolutions.
- Developed methods for changepoint analysis in multiple shape-based applications.

- Visualized & edit geospatial data using ArcGIS, QGIS, & Python (GeoPandas, ArcPy).
 - Collaborated with professionals across a variety of scientific fields including climate, ecology, computer science, biology, & a variety of federal agencies.
 - Traveled to Antarctica to perform fieldwork & work on ships.
- National Institute of Standards and Technology**, Gaithersburg, MD
Guest Researcher May 2022 - June 2024
 - Collaborated with imaging scientists on current Ph.D. research.
 - Implemented image analysis & computer vision techniques in Python & MATLAB.
 - Developed new algorithms for shape processing & statistical analysis.
- Quark Expeditions**, Ushuaia, Tierra del Fuego, Argentina
Research Scientist with PenguinWatch November 2022 - December 2022
 - Collaborated with Dr. Tom Hart from Oxford University on penguin colony research.
 - Collected guano samples, flew drones, & upkept trail cameras.
 - Gave lectures & educated ship passengers on Antarctic ecology.
 - Assisted with setting up campsites, biosecurity & disinfection, safety drills.
- Aptima**, Woburn, MA
Data Scientist Intern June 2021 - August 2021
 - Implemented methods for studying motion & biometric data in Python.
 - Performed speech recognition & sentiment analysis in Python.
 - Researched social network analysis methods to study team cooperation.
 - Analyzed large & noisy datasets.
- Fox Chase Cancer Center**, Philadelphia, PA
Research Assistant to Dr. Andrew J. Andrews June 2020 - August 2020
 - Modeled & simulated enzyme-catalyzed reactions using MATLAB/Python.
 - Analyzed & interpreted data using statistical & machine learning methods (SVM classification/regression, PCA).
- University of Minnesota**, Minneapolis, MN
Teaching Assistant to Dr. Duane Nykamp September 2019 - December 2019
 - Taught students to model biological processes using dynamical systems.
 - Aided students in understanding calculus concepts applied to the life sciences.
 - Helped students with homework assignments & project creation.
- Fox Chase Cancer Center**, Philadelphia, PA
Undergraduate Research Fellow to Dr. Andrew J. Andrews June 2019 - August 2019
 - Simulated enzyme-catalyzed reactions in Python.
 - Researched machine learning methods such as SVM, PCA, & regression.
 - Wrote progress reports, created presentations.

- **Rose: Smarter Mental Health**, New York, NY

Machine Learning Intern

September 2018 - May 2019

- Assisted in building a healthcare tech start-up from the ground up.
- Helped value the company at over \$2 million during employment.
- Built machine & deep learning models for sentiment analysis (tokenization, contextual embedding, speech tagging) & large dataset analysis using MATLAB & Python.
- Analyzed & prepared large sets of labeled & unlabeled data.
- Used AWS for database creation & maintenance.

- **University of Minnesota**, Minneapolis, MN

Teaching Assistant in the Math Center for Educational Programs

September 2018 - May 2020

- Taught advanced mathematics topics to middle & high school students during the week.
- Ran weekend activities to engage students in the community to applications of mathematics.
- Graded homework & exams, prepared paperwork, & interacted with parents.

- **University of Minnesota**, Minneapolis, MN

Undergraduate Researcher in Combinatorics/Number Theory

January 2018 - December 2019

- Prepared weekly presentations, attended conferences, & recorded work in publications.

ACTIVITIES & SOCIETIES

- **SIAM Student Chapter**, Stony Brook University

October 2022 - August 2025

President

- **International Pals Program**, Stony Brook University

September 2023 - August 2024

Mentor

- **Heather Lynch Lab**, Stony Brook University

June 2023 - August 2024

Research Mentor

- **Undergraduate Research Journal**, University of Minnesota

September 2018 - May 2020

Managing Editor, Treasurer

HONORS & AWARDS

- Institute for Advanced Computational Science (IACS) Junior Researcher Fellowship Award Renewal (2023-2024).
- Society for Industrial and Applied Mathematics (SIAM) Award for Outstanding Service in Development of the Stony Brook Student Chapter (2023).
- Stony Brook University Applied Mathematics and Statistics Department Special Award for Outstanding Service (2023).
- Society for Industrial and Applied Mathematics (SIAM) Travel Award (2023).
- Institute for Advanced Computational Science (IACS) Junior Researcher Fellowship Award (2022-2023).
- Society for Industrial and Applied Mathematics (SIAM) Travel Award (2022).
- Mathematical Association of American Outstanding Poster Award (2019).

PUBLICATIONS, IN-REVIEW, & APPEARANCES:

- Hall, Carole et al. (2025- in review) "A molecular dynamics-based approach to the recovery of spatial information from low-resolution imagery of penguin colonies". *Physical Review E*.
- Gonzalez, Efrain & Hall, Carole et al. (2025 - Accepted). "Dynamic Mode Decomposition in Subspace Tracking and Object Detection". *IEEE Aerospace*.
- Hall, Carole & Tucker, J. Derek. (2025 July). " From Shapes to Shifts: An Elastic Framework for Detecting Changepoints in Animal Movement". Talk presented at the Society for Industrial and Applied Mathematics (SIAM) Conference on Computational Geometric Design (GD25). Montreal, Québec, Canada.
- Wu, Haoyu & Hall, Carole et al. (2024). "Penguin colony georegistration using camera pose estimation and phototourism". *PLOS ONE*. 19. 10.1371/journal.pone.0311038.
- Hall, Carole. (2023 June). "From Pixels to Penguins: Analyzing Long-term Adélie Colony Dynamics Through Super-resolution of Satellite Imagery". Talk presented at the POLE Seminar at Woods Hole Oceanographic Institute, Woods Hole, Massachusetts.
- Hall, Carole, et al. (2023 May). "From Pixels to Penguins: Analyzing Long-term Adélie Colony Dynamics Using Landsat Imagery". Talk presented at the NASA Carbon Cycles & Ecosystems Joint Science Workshop, College Park, Maryland.
- Hall, Carole. (2023 February). "The Physics of Penguins: Super-Resolving Imagery of Adélie Penguin Colonies". Talk presented at the Society for Industrial and Applied Mathematics (SIAM) Computational Science and Engineering Conference, Amsterdam, The Netherlands.
- Hall, Carole. (2022 July). "Modeling the Spatial Aggregation of Adélie Penguins". Talk presented at the Society for Industrial and Applied Mathematics (SIAM) Mathematics of Planet Earth Conference, Pittsburgh, PA.
- Hall, Carole. (2022 March). "How the complexity of Adélie penguin colony shape relates to the chance of colony collapse". Talk presented at the Aspen Center for Physics Conference on the Dynamics of Social Interactions, Aspen, CO.
- Adams, Ashleigh & Hall, Carole. (2019, January). "Simplicial Complexes of Zero-Sumfree Sets". Poster session presented at the Mathematics Association of America Undergraduate Poster Session, Baltimore, MD.
- Adams, Ashleigh & Hall, Carole. (2018, October). "Simplicial Complexes of Zero-Sumfree Sets". Talk presented at the Midwest Conference on Combinatorics and Combinatorial Computing, Duluth, MN.
- Adams, Ashleigh & Hall, Carole. (2018, August). "Simplicial Complexes of Zero-Sumfree Sets". Poster session presented at the 2018 Summer Undergraduate Research Symposium, Minneapolis, MN.
- Adams, A., Hall, C., & Stucky, E. (2019). Classifications of ℓ -Zero-Sumfree Sets. *The PUMP Journal of Undergraduate Research*, 2, 179-198. Retrieved from <https://journals.calstate.edu/pump/article/view/1805>.